A COMPLEX PERFECT MODULO A NILPOTENT IDEAL IS PERFECT

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- A complex of projective R-modules representing K whose reduction modulo I is a given complex lifting the representative of the derived quotient exists. invoked at proof lines 20523-20525. Provides the projective lifting P of K with P/IP the chosen finite complex.
- Nakayama's lemma: a finitely generated module over R whose reduction modulo a nilpotent ideal is finite projective over R/I is finite projective over R. invoked at proof lines 20526-20527. Concludes each P^n is finite projective over R.

Source reference. Original definition 0657: sources/stacks-more-algebra/pinned-original-source.tex, source lines 18871–18871. Cited at proof lines 20519-20520 for the shape of the chosen representative; the definition is used inside the counted span.

Source reference. Original lemma 09AR: sources/stacks-more-algebra/pinned-original-source.tex, source lines 19469–19469. Cited at proof lines 20523-20525 to produce the projective lifting P; the conclusion is used inside the counted span.

Source reference. Original lemma 00DV: sources/stacks-chow/tags.txt, source lines 512–512. Cited at proof lines 20526-20527 to conclude the lifting is termwise finite projective; the conclusion is used inside the counted span.

Lemma 0.2. *Let R be a ring. Let $I \subset R$ be an ideal. Let K be an object of $D(R)$. Assume that*

(1) *$K \otimes^{\mathbf{L}}_R R/I$ is perfect in $D(R/I)$, and*
(2) *I is a nilpotent ideal.*

Then K is perfect in $D(R)$.

Proof. Choose a finite complex $\overline{P}^\bullet$ of finite projective R/I-modules representing $K \otimes^{\mathbf{L}}_R R/I$, see Definition 0657. By Lemma 09AR there exists a complex $P^\bullet$ of projective R-modules representing K such that $\overline{P}^\bullet = P^\bullet/IP^\bullet$. It follows from Nakayama's lemma (Algebra, Lemma 00DV) that $P^\bullet$ is a finite complex of finite projective R-modules. □

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